Performance benchmarks using the Rust Scientific Library (Russell)

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We test the linear system $\boldsymbol{A}\,\boldsymbol{x} = \boldsymbol{b}$ where $\boldsymbol{A}$ is the coefficient matrix, $\boldsymbol{x}$ is the solution vector, and $\boldsymbol{b}$ is the right-hand side vector. The coefficient matrix is set with matrices from the SuiteSparse Matrix Collection. The right-hand side vector is filled with ones, i.e., we study the solution of

$$\boldsymbol{A}\,\boldsymbol{x} = \mathbf{1} \tag{1}$$

The relative error is calculated as

$$\texttt{RelativeError} = \frac{\max(|\boldsymbol{A}\,\boldsymbol{x} - \mathbf{1}|)}{\max(|\boldsymbol{A}| + 1)} \tag{2}$$

The tested real-valued matrices are:

1 `bwm2000` (Bai) – Brusselator wave model in transport interaction of chemical solutions (1992)
2 `rdb5000` (Bai) – Reaction-diffusion brusselator model (1994)
3 `Goodwin_040` (Goodwin) – Finite element, Navier-Stokes & other transport equations (2018)
4 `fp` (MKS) – 2-D Fokker Planck eqn, electron dyn. in external field (2006)
5 `xenon1` (Ronis) – Complex zeolite, sodalite crystals (2001)
6 `twotone` (ATandT) – Harmonic balance method (2001)
7 `Raj1` (Rajat) – Circuit Simulation Problem (2007)
8 `boyd2` (GHS_indef) – Optimization problem (2004)
9 `Goodwin_071` (Goodwin) – Finite element, Navier-Stokes & other transport equations (2018)
10 `darcy003` (GHS_indef) – Discretization using mixed FE of Darcy (2002)
11 `rma10` (Bova) – 3D CFD model, Charleston harbor (1997)
12 `helm2d03` (GHS_indef) – Helmholtz eq on a unit square (2004)
13 `stomach` (Norris) – 3D electro-physical model of a duodenum (2003)
14 `oilpan` (GHS_psdef) – Structural problem (2004)
15 `ASIC_680k` (Sandia) – Circuit simulation matrix (2006)
16 `tmt_unsym` (CEMW) – Electromagnetics problem (2008)
17 `Goodwin_127` (Goodwin) – Finite element, Navier-Stokes & other transport equations (2018)
18 `pre2` (ATandT) – Harmonic balance method (2001)
19 `marine1` (Martin) – Chemical oceanography; a marine nitrogen cycle inverse model (2018)
20 `torso1` (Norris) – Finite differences and boundary element, 2D model of torso (2003)
21 `atmosmodd` (Bourchtein) – CFD analysis of atmospheric models (2009)
22 `atmosmodl` (Bourchtein) – CFD analysis of atmospheric models (2009)
23 `memchip` (Freescale) – Circuit simulation problem (2010)
24 `Freescale1` (Freescale) – Circuit simulation problem (2008)
25 `rajat31` (Rajat) – Circuit simulation problem (2006)
26 `Transport` (Janna) – 3D finite element flow and transport (2012)
27 `inline_1` (GHS_psdef) – Structural problem, stiffness matrix (2004)
28 `PFlow_742` (Janna) – 3D pressure-temperature evolution in porous media (2014)
29 `Emilia_923` (Janna) – Geomechanical model for C02 sequestration (2011)
30 `dielFilterV2real` (Dziekonski) – FEM in electromagnetics (2011)
31 `Flan_1565` (Janna) – Structural problem, 3D model of a steel flange (2011)
32 `pres-cylin` (Pedroso) – FEM stiffness matrix of a pressurized cylinder (Tet10 with 1,711,464 DOF). Not from the Collection. From Pedroso DM (2024) Caveats of three direct linear solvers for finite element analyses.

The tested complex-valued matrices are:

1 `mhd1280b` (Bai) – Alfven spectra in magnetohydrodynamics (1994)

2. `mplate` (Cote) – Vibro-acoustic problem (1997)
3. `RFdevice` (Rost) – Semiconductor device simulation (2007)
4. `vfem` (CEMW) – Electromagnetics; vector finite element (2008)
5. `fem_filter` (Lee) – FEM band-pass microwave filter 500MHz (2008)
6. `Chevron4` (Chevron) – Temporal freq domain seismic modeling (2012)
7. `mono_500Hz` (FreeFieldTechnologies) – 3D vibro-acoustic problem, aircraft engine nacelle (2008)
8. `kim2` (Kim) – 2D 676-by-676 complex mesh (2002)
9. `fem_hifreq_circuit` (Lee) – FEM Maxwell equations for hi-freq circuit (2009)
10. `dielFilterV3clx` (Dziekonski) – High-order vector finite element method in EM (2011)

Additional notes:

- Each problem is solved ten times. The reported error is the maximum among runs. The reported computer time is the average without outliers.
- Total time includes initialization (memory allocation + symbolic factorization), numeric factorization, and solve.
- Column NNZ (number of non-zeros) corresponds to Pattern Entries reported by the SuiteSparse Matrix Collection.
- Column Sym shows if symmetry information was provided to the solver. An asterisk means positive definite information was also given.
- oom (out-of-memory) indicates that the symbolic factorization was terminated due to insufficient memory.
- Bold values highlight significant results, such as large errors or high computation times.
- The hybrid memory mode is enabled for the pres-cylin matrix and cuDSS.

Information about the system:

```
--- OS ---
NAME="Arch Linux"
KERNEL=7.0.9-arch2-1

--- GPU ---
GPU[0]: NVIDIA GeForce RTX 4090, 595.71.05, 24564 MiB

--- CPU ---
Architecture       : x86_64
CPU(s)             : 32
On-line CPU(s) list: 0-31
Model name         : 13th Gen Intel(R) Core(TM) i9-13900KF
Thread(s) per core : 2
Core(s) per socket : 24
Socket(s)          : 1
CPU(s) scaling MHz : 24%
CPU max MHz        : 5800.0000
CPU min MHz        : 800.0000
BogoMIPS           : 5990.40
L1d cache          : 896 KiB (24 instances)
L1i cache          : 1.3 MiB (24 instances)
L2 cache           : 32 MiB (12 instances)
L3 cache           : 36 MiB (1 instance)
NUMA node0 CPU(s)  : 0-31
Vulnerability L1tf : Not affected

--- Memory ---
MemTotal:       32616240 kB
MemFree:        17676884 kB
MemAvailable:   27546872 kB
SwapTotal:      36806824 kB
```

Information about the libraries:

```
cuDSS: 0.8.0.10 (CUDA 13)
MUMPS: 5.9.0
SuiteSparse: latest (from GitHub)
```

The results are given in Tables 1, 2, 3, and 4.

Table 1: Calculations on Arch. cuDSS and MUMPS. Real-valued matrices.

Matrix	Nrow	NNZ	Sym	cuDSS Matching Algorithm	cuDSS Total Time	cuDSS Relative Error	MUMPS Total Time	MUMPS Relative Error
bwm2000	2,000	7,996	No	Auto	8.633ms	8.21×10^{-16}	6.260ms	7.50×10^{-16}
rdb5000	5,000	29,600	No	Auto	30.087ms	4.06×10^{-13}	7.442ms	5.37×10^{-13}
Goodwin_040	17,922	561,677	No	MaxMinDiag	71.095ms	2.82×10^{-12}	106.490ms	1.65×10^{-11}
fp	7,548	848,553	No	Auto	155.376ms	4.98×10^{-9}	164.943ms	1.16×10^{-20}
xenon1	48,600	1,181,120	No	Auto	194.980ms	1.32×10^{-39}	308.937ms	6.74×10^{-40}
twotone	120,750	1,224,224	No	Auto	772.871ms	4.73×10^{-13}	879.251ms	1.93×10^{-13}
Raj1	263,743	1,302,464	No	Auto	883.834ms	1.21×10^{-11}	901.208ms	3.63×10^{-13}
boyd2	466,316	1,500,397	Yes	None	996.540ms	3.24×10^{-11}	755.545ms	8.98×10^{-11}
Goodwin_071	56,021	1,797,934	No	MaxMinDiag	174.813ms	8.65×10^{-12}	258.507ms	8.38×10^{-11}
darcy003	389,874	2,101,242	Yes	MaxMinDiagAlt	2.438s	5.64×10^{-7}	10.667s	1.75×10^{-10}
rma10	46,835	2,374,001	No	Auto	241.711ms	9.04×10^{-16}	150.619ms	1.35×10^{-16}
helm2d03	392,257	2,741,935	Yes	None	1.239s	1.68×10^{-10}	1.440s	3.82×10^{-10}
stomach	213,360	3,021,648	No	Auto	1.280s	2.71×10^{-15}	1.579s	1.49×10^{-15}
oilpan	73,752	3,597,188	Yes*	None	102.687ms	4.65×10^{-15}	185.724ms	2.22×10^{-15}
ASIC_680k	682,862	3,871,773	No	Auto	3.458s	7.33×10^{-11}	**1m17.353s**	8.09×10^{-11}
tmt_unsym	917,825	4,584,801	No	Auto	2.569s	4.95×10^{-7}	3.061s	1.43×10^{-7}
Goodwin_127	178,437	5,778,545	No	MaxMinDiag	372.312ms	6.83×10^{-11}	846.998ms	6.20×10^{-10}
pre2	659,033	5,959,282	No	Auto	3.428s	2.15×10^{-14}	4.383s	2.60×10^{-14}
marine1	400,320	6,226,538	No	Auto	4.032s	1.70×10^{-8}	5.352s	6.25×10^{-14}
torso1	116,158	8,516,500	No	MaxDiagCount	1.050s	5.90×10^{-7}	1.847s	$\mathbf{3.75\times10^{-6}}$
atmosmodd	1,270,432	8,814,880	No	Auto	18.668s	1.44×10^{-16}	42.718s	2.48×10^{-16}
atmosmodl	1,489,752	10,319,760	No	Auto	18.165s	9.34×10^{-18}	38.733s	7.60×10^{-18}
memchip	2,707,524	14,810,202	No	Auto	6.941s	3.23×10^{-15}	7.546s	3.40×10^{-15}
Freescale1	3,428,755	18,920,347	No	Auto	8.076s	9.39×10^{-10}	9.580s	7.05×10^{-10}
rajat31	4,690,002	20,316,253	No	Auto	17.285s	3.26×10^{-14}	15.175s	8.73×10^{-15}
Transport	1,602,111	23,500,731	No	Auto	21.417s	7.21×10^{-10}	41.188s	3.81×10^{-10}
inline_1	503,712	36,816,342	Yes*	None	1.762s	5.86×10^{-15}	3.261s	3.38×10^{-15}
PFlow_742	742,793	37,138,461	Yes*	None	9.792s	1.77×10^{-10}	10.706s	4.17×10^{-11}
Emilia_923	923,136	41,005,206	Yes*	None	15.903s	1.50×10^{-22}	37.580s	5.27×10^{-23}
dielFilterV2	1,157,456	48,538,952	Yes	None	7.007s	1.38×10^{-11}	12.388s	1.49×10^{-11}
Flan_1565	1,564,794	117,406,044	Yes*	None	10.212s	2.24×10^{-15}	19.735s	9.98×10^{-16}
pres-cylin	1,711,464	133,562,188	Yes*	None	**2m4.270s**	8.89×10^{-13}	**1m18.368s**	5.09×10^{-13}

Table 2: Calculations on Arch. cuDSS and MUMPS. Complex-valued matrices.

				cuDSS			MUMPS	
Matrix	**Nrow**	**NNZ**	**Sym**	**Matching Algorithm**	**Total Time**	**Relative Error**	**Total Time**	**Relative Error**
mhd1280b	1,280	12,029	No	Auto	7.281ms	1.57×10^{-15}	4.551ms	1.17×10^{-15}
mplate	5,962	142,190	Yes	None	78.587ms	1.13×10^{-10}	167.040ms	6.28×10^{-11}
RFdevice	74,104	365,580	No	MaxDiagSum	570.478ms	$\mathbf{3.21\times10^{-2}}$	1.548s	1.56×10^{-7}
vfem	93,476	1,434,636	No	Auto	865.606ms	5.29×10^{-8}	1.403s	8.74×10^{-8}
fem_filter	74,062	1,731,206	No	Auto	713.189ms	1.81×10^{-10}	1.039s	1.44×10^{-10}
Chevron4	711,450	6,376,412	No	Auto	2.480s	8.51×10^{-11}	3.139s	4.16×10^{-11}
mono_500Hz	169,410	5,036,288	No	Auto	2.735s	2.23×10^{-10}	4.830s	5.99×10^{-9}
kim2	456,976	11,330,020	No	Auto	13.916s	0.00×10^{0}	4.346s	2.73×10^{-18}
fem_hifreq_circuit	491,100	20,239,237	No	Auto	4.871s	1.60×10^{-10}	9.018s	1.28×10^{-10}
dielFilterV3clx	420,408	32,886,208	Yes	None	2.184s	4.31×10^{-10}	3.969s	1.73×10^{-11}

Table 3: Calculations on Arch. UMFPACK and MUMPS. Real-valued matrices.

Matrix	Nrow	NNZ	Sym	UMFPACK Total Time	UMFPACK Relative Error	MUMPS Total Time	MUMPS Relative Error
bwm2000	2,000	7,996	No	713.627µs	5.30×10^{-16}	6.260ms	7.50×10^{-16}
rdb5000	5,000	29,600	No	11.504ms	1.93×10^{-15}	7.442ms	5.37×10^{-13}
Goodwin_040	17,922	561,677	No	104.256ms	6.62×10^{-13}	106.490ms	1.65×10^{-11}
fp	7,548	848,553	No	149.956ms	2.57×10^{-21}	164.943ms	1.16×10^{-20}
xenon1	48,600	1,181,120	No	582.982ms	5.55×10^{-40}	308.937ms	6.74×10^{-40}
twotone	120,750	1,224,224	No	334.710ms	1.13×10^{-14}	879.251ms	1.93×10^{-13}
Raj1	263,743	1,302,464	No	oom	oom	901.208ms	3.63×10^{-13}
boyd2	466,316	1,500,397	Yes	**1m19.604s**	2.86×10^{-13}	755.545ms	8.98×10^{-11}
Goodwin_071	56,021	1,797,934	No	418.565ms	2.77×10^{-12}	258.507ms	8.38×10^{-11}
darcy003	389,874	2,101,242	Yes	1.072s	6.95×10^{-12}	10.667s	1.75×10^{-10}
rma10	46,835	2,374,001	No	398.716ms	1.57×10^{-17}	150.619ms	1.35×10^{-16}
helm2d03	392,257	2,741,935	Yes	1.375s	2.37×10^{-13}	1.440s	3.82×10^{-10}
stomach	213,360	3,021,648	No	1.681s	5.13×10^{-16}	1.579s	1.49×10^{-15}
oilpan	73,752	3,597,188	Yes*	481.117ms	1.26×10^{-15}	185.724ms	2.22×10^{-15}
ASIC_680k	682,862	3,871,773	No	1.242s	2.20×10^{-11}	**1m17.353s**	8.09×10^{-11}
tmt_unsym	917,825	4,584,801	No	3.655s	5.96×10^{-8}	3.061s	1.43×10^{-7}
Goodwin_127	178,437	5,778,545	No	1.380s	6.03×10^{-12}	846.998ms	6.20×10^{-10}
pre2	659,033	5,959,282	No	oom	oom	4.383s	2.60×10^{-14}
marine1	400,320	6,226,538	No	oom	oom	5.352s	6.25×10^{-14}
torso1	116,158	8,516,500	No	1.314s	3.82×10^{-8}	1.847s	$\mathbf{3.75\times10^{-6}}$
atmosmodd	1,270,432	8,814,880	No	oom	oom	42.718s	2.48×10^{-16}
atmosmodl	1,489,752	10,319,760	No	oom	oom	38.733s	7.60×10^{-18}
memchip	2,707,524	14,810,202	No	28.947s	2.04×10^{-15}	7.546s	3.40×10^{-15}
Freescale1	3,428,755	18,920,347	No	28.693s	7.05×10^{-10}	9.580s	7.05×10^{-10}
rajat31	4,690,002	20,316,253	No	oom	oom	15.175s	8.73×10^{-15}
Transport	1,602,111	23,500,731	No	oom	oom	41.188s	3.81×10^{-10}
inline_1	503,712	36,816,342	Yes*	oom	oom	3.261s	3.38×10^{-15}
PFlow_742	742,793	37,138,461	Yes*	oom	oom	10.706s	4.17×10^{-11}
Emilia_923	923,136	41,005,206	Yes*	oom	oom	37.580s	5.27×10^{-23}
dielFilterV2	1,157,456	48,538,952	Yes	oom	oom	12.388s	1.49×10^{-11}
Flan_1565	1,564,794	117,406,044	Yes*	oom	oom	19.735s	9.98×10^{-16}
pres-cylin	1,711,464	133,562,188	Yes*	oom	oom	**1m18.368s**	5.09×10^{-13}

Table 4: Calculations on Arch. UMFPACK and MUMPS. Complex-valued matrices.

Matrix	Nrow	NNZ	Sym	UMFPACK Total Time	UMFPACK Relative Error	MUMPS Total Time	MUMPS Relative Error
mhd1280b	1,280	12,029	No	345.764μs	7.41×10^{-16}	4.551ms	1.17×10^{-15}
mplate	5,962	142,190	Yes	412.395ms	2.32×10^{-11}	167.040ms	6.28×10^{-11}
RFdevice	74,104	365,580	No	3.901s	9.16×10^{-14}	1.548s	1.56×10^{-7}
vfem	93,476	1,434,636	No	5.132s	2.07×10^{-8}	1.403s	8.74×10^{-8}
fem_filter	74,062	1,731,206	No	2.366s	6.32×10^{-11}	1.039s	1.44×10^{-10}
Chevron4	711,450	6,376,412	No	4.528s	1.45×10^{-11}	3.139s	4.16×10^{-11}
mono_500Hz	169,410	5,036,288	No	oom	oom	4.830s	5.99×10^{-9}
kim2	456,976	11,330,020	No	oom	oom	4.346s	2.73×10^{-18}
fem_hifreq_circuit	491,100	20,239,237	No	oom	oom	9.018s	1.28×10^{-10}
dielFilterV3clx	420,408	32,886,208	Yes	oom	oom	3.969s	1.73×10^{-11}